MT4013 - Material Sensor

Metal Oxide Nanostructures in Sensing

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1. Introduction

- Polusi udara: sulfur dioksida (SO_2), karbon monoksida (CO), dan nitrogen dioksida (NO_2)
- Juga partikel padat dan cair (terutama logam berat) tersuspensi di udara

- Diperlukan deteksi dini
- Saat ini, sensor gas berbasis pada oksida logam semikonduktor dan polimer.

1. Introduction

Advantages:

- Low cost
- Size
- Simple measuring electronics
- Morphology effect

Disadvantages:

- selectivity
- long-term drift: stoichiometry changes and coalescence (penggabungan) of nanocrystallites

2. Basic Sensing Characteristics

1) Sensor Response
2) Response Time
3) Recovery Time
4) Operating Temperature
5) Selectivity
6) Detection Limit
7) Stability and Repeatability

Sensor Response

Sensor response can be specified as the ratio of some physical parameters/signals:

- resistance,
- conductance,
- photoluminescence,
- reflectance, etc.

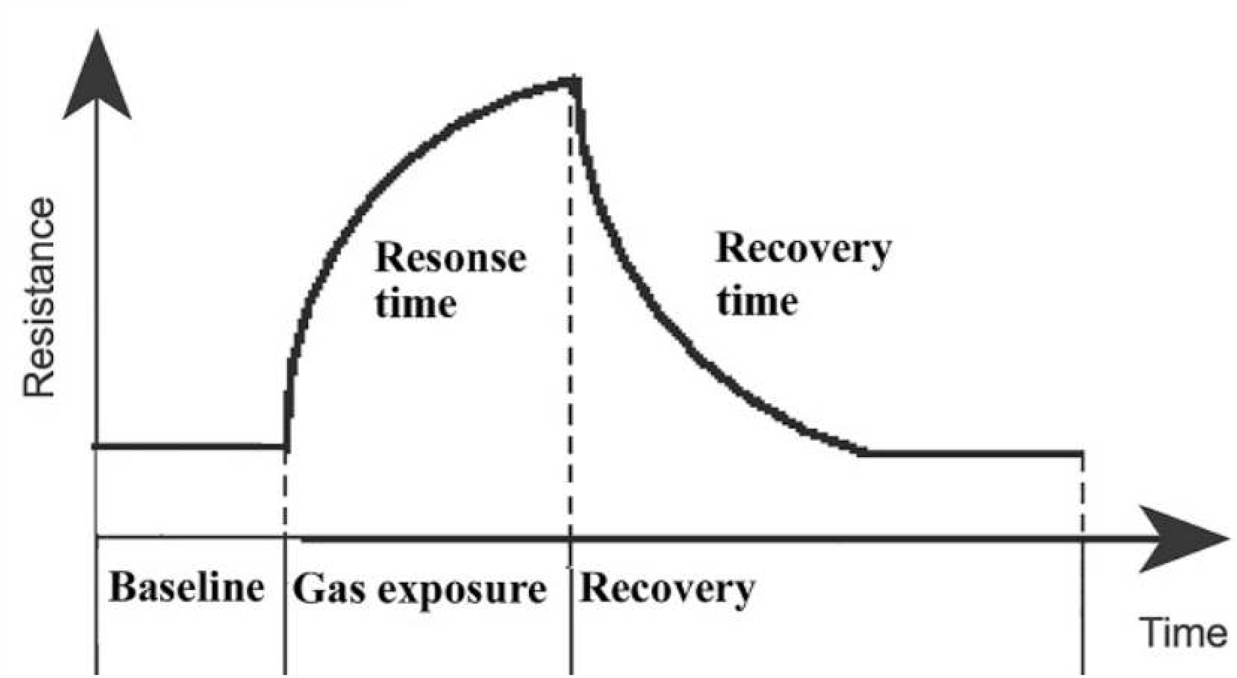
Gas resistance Initial resistance

$$S = \frac{R_g - R_0}{R_0} = \frac{\Delta R}{R_0}$$

The change of the sensor parameter before and after the target molecules exposure.

Response & Recovery Time

Response time is the time that a sensor requires to change its signal (e.g., resistance) from the initial state (baseline) to a final value (~90% of the final value).



Sensor response curve vs recovery time

Recovery time is the time to return the measured signal to the baseline value after the target molecules are removed (to 90% of the original base-line signal).

Operating Temperature

- Optimum operating temperature depends on the material of the sensor and the kind of gases.

Selectivity

- The selectivity of sensors refers to the characteristics that define whether the sensor can respond selectively to a particular target analyte in the presence of other analytes under similar operating conditions.

$$\text{Selectivity} = \frac{S_i}{S_g}$$

sensitivity of the sensor toward interface species

sensitivity of the sensor toward target species

Detection Limit

- The **limit of detection (LOD)** is the minimum concentration of the target molecules, which can be detected by a sensor under certain conditions.

Stability and Repeatability

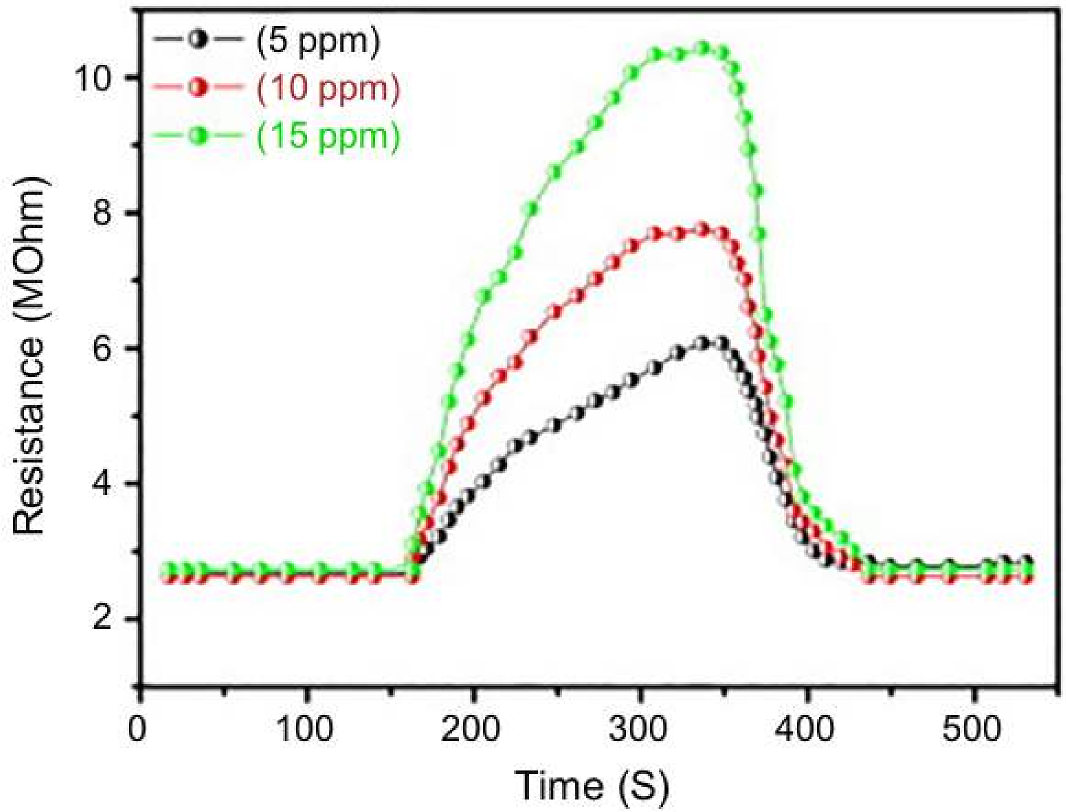
- The **stability/repeatability** of the sensor material determines the possibility of multiple recoveries of its parameters after many cycles of sensor performance.

3. Metal Oxides Nanostructures for Different Gas Sensing

1) Oxygen Sensors
2) CO Sensors
3) NO, N_2O, and NO_2 Sensors
4) VOCs sensors

Oxygen Sensors

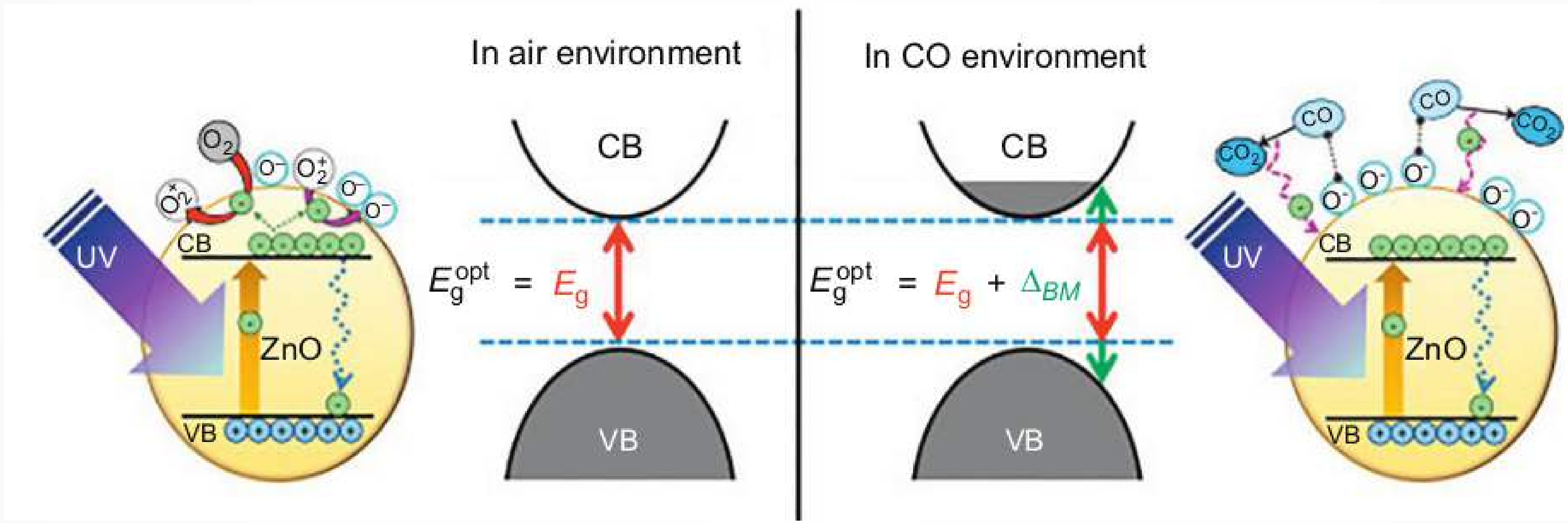
- Mostly, the sensors based on metal oxide nanostructures offer advantages in high sensitivity and selectivity, easy fabrication, and low production cost.
- SnO_2
- TiO_2
- ZnO
- $SrTiO_3$
- WO_3



Sensor response of Mn-doped ZnO nanorod gas sensor for different oxygen concentrations RT

CO Sensors

- CO is a colorless and poisonous gas produced by incomplete burning of carbon-based fuels



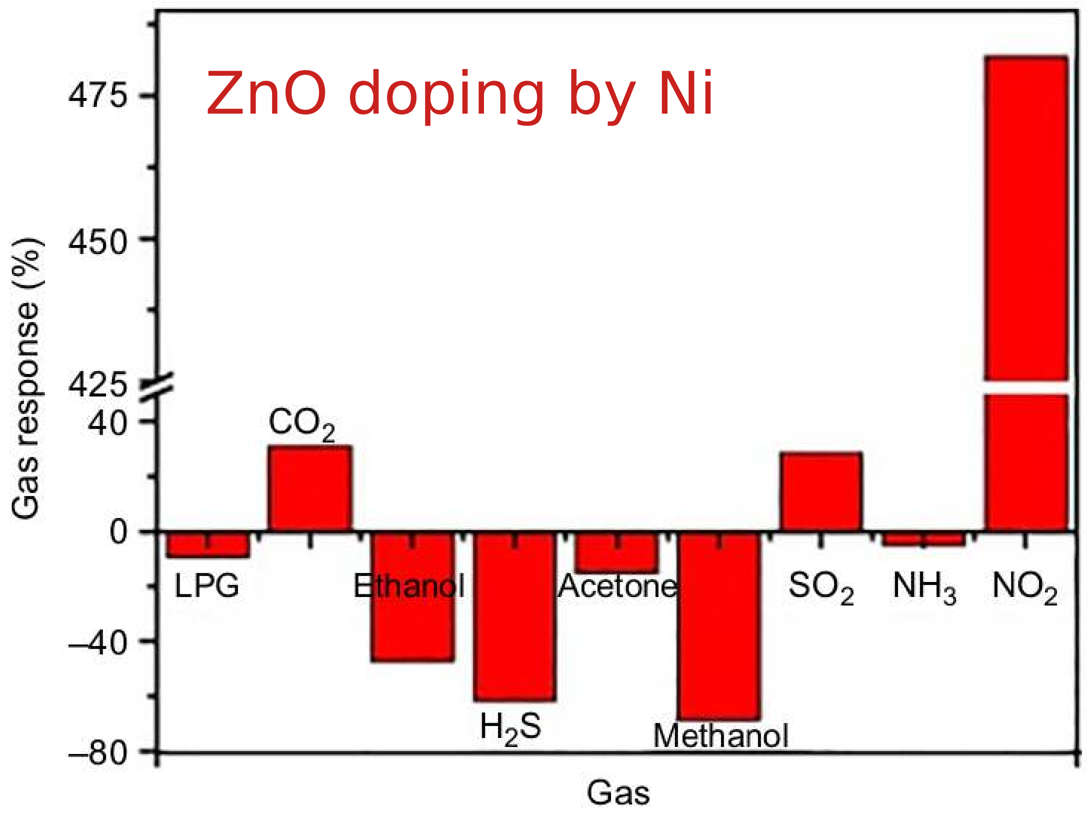
- UV illumination ⇒ the electrons in ZnO are excited to the CB giving rise to higher carrier concentration in CB ⇒ increased the adsorption of oxygen.
- CO molecules interact with the adsorbed oxygen species on ZnO surface to form CO_2 ⇒ additional free electron is moving into the CB, which enhances the attaching of CO molecules to the surface improving the gas sensing properties

NO, N_2O, and NO_2 Sensors

- NO_x gases are the most reactive and represent the highest risk to human health and natural environment.

side products in industrial chemical processes:

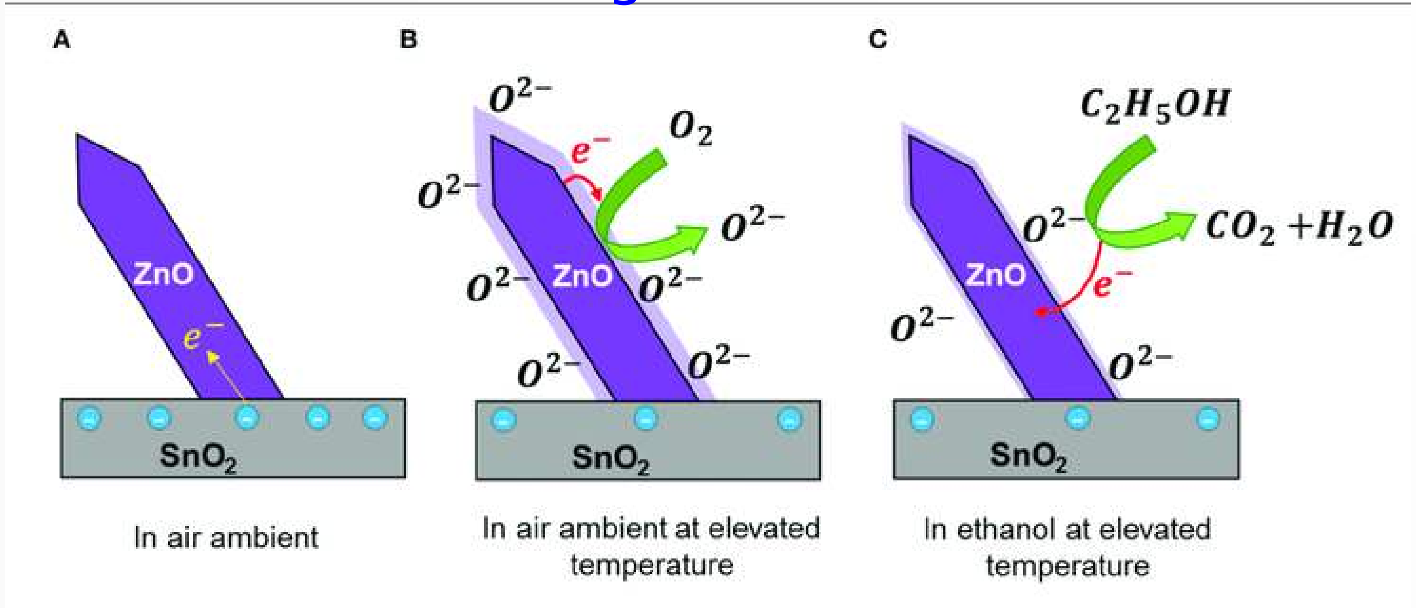
- power plants fuels burning,
- heavy equipment used for construction works, cars, trucks, and autobuses
- agricultural activity.

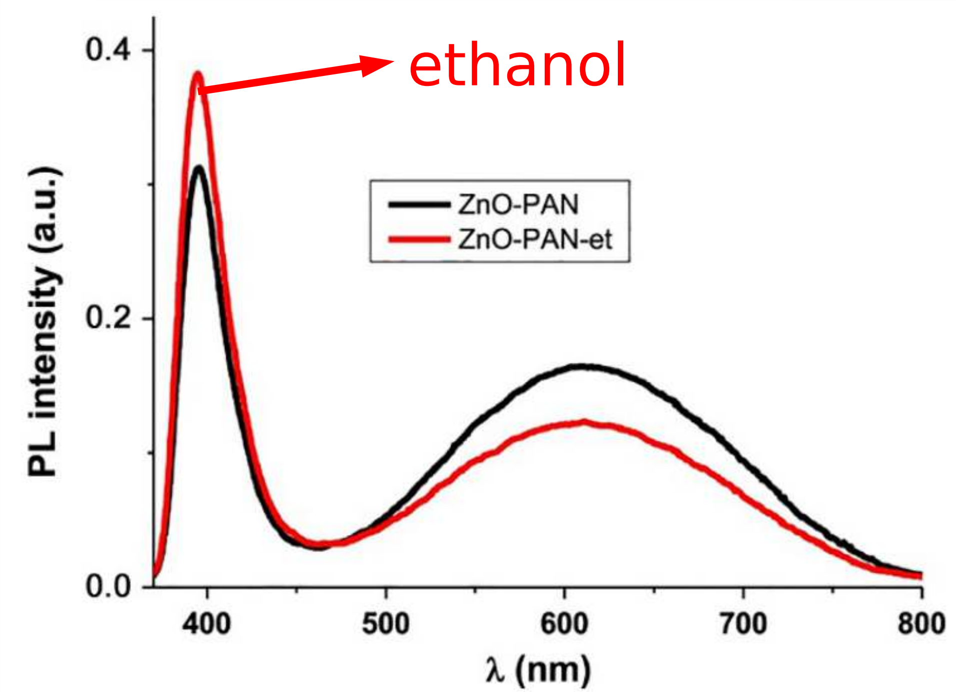


VOCs sensors

- Volatile organic compounds (VOCs) are organic chemicals that have a high vapor pressure at ordinary room temperature ⇒ benzene, toluene, ethyl-benzene, acetone, ethanol, etc.

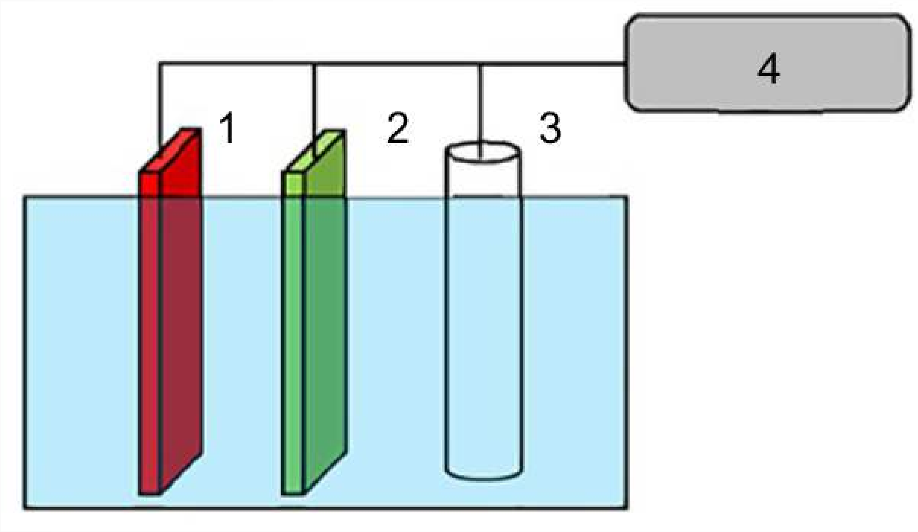
Sensing mechanism

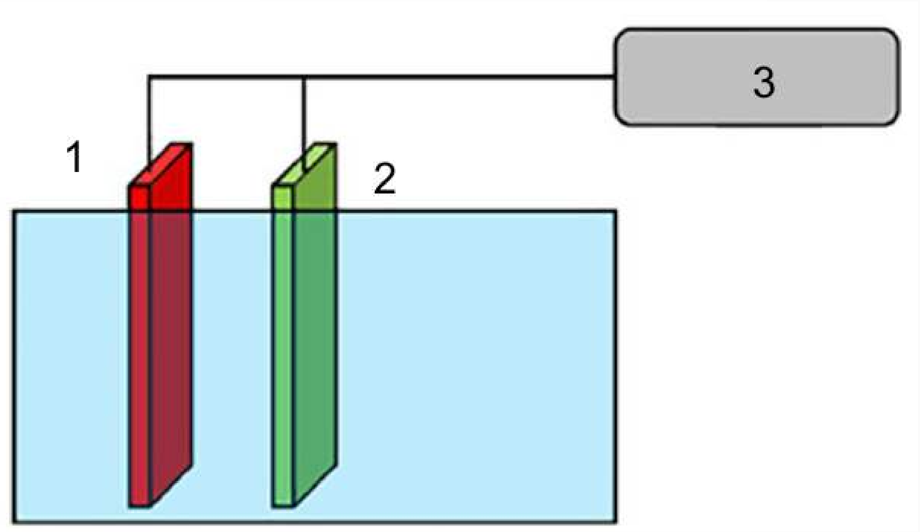




4. Electrochemical Sensors

- electrochemical sensors work on the charge transfer principle, resulted from a reduction/oxidation reaction between target molecules and the electrode surface.

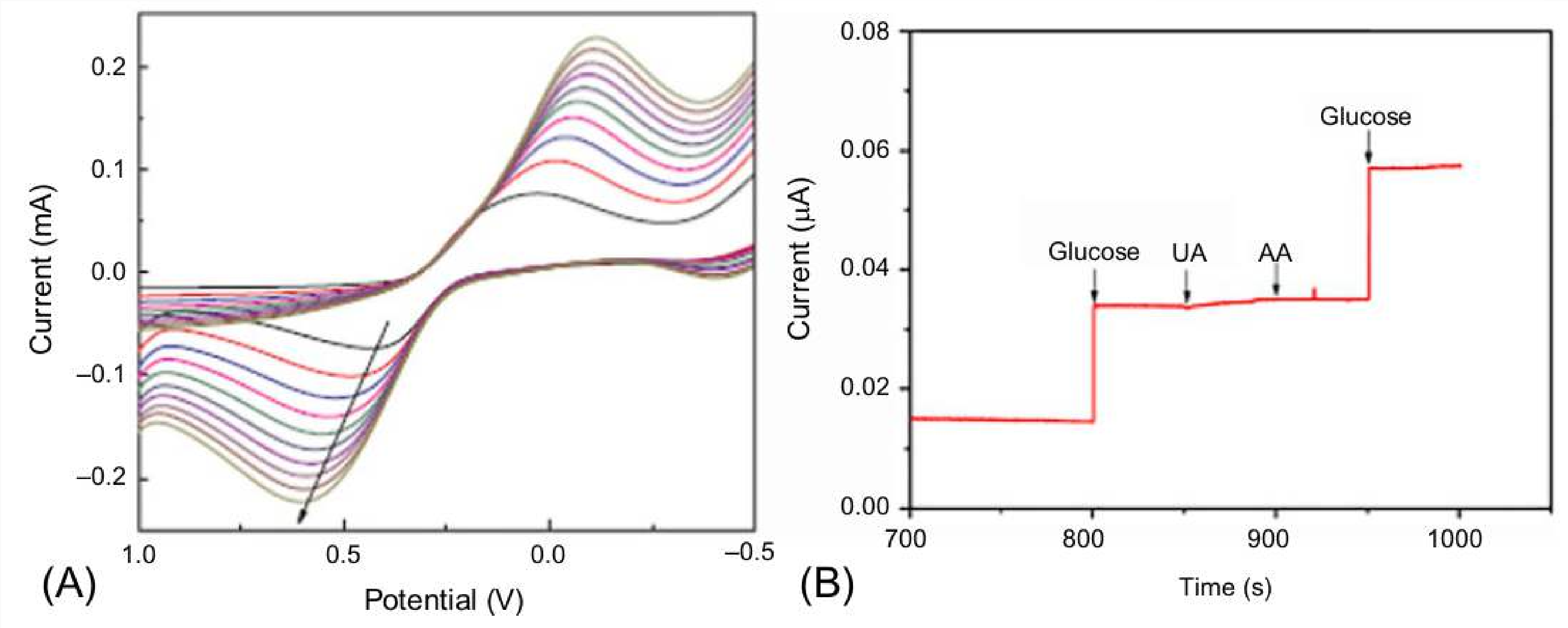




- Schematic representation of three electrode and two electrode electrochemical measurements.
- The three-electrode scheme consists of a working electrode (sensor), a counter electrode (typically platinum), a reference electrode 3 (Ag/AgCl electrode in 3 M potassium chloride solution), and electrochemical analyzer.

4. Electrochemical Sensors

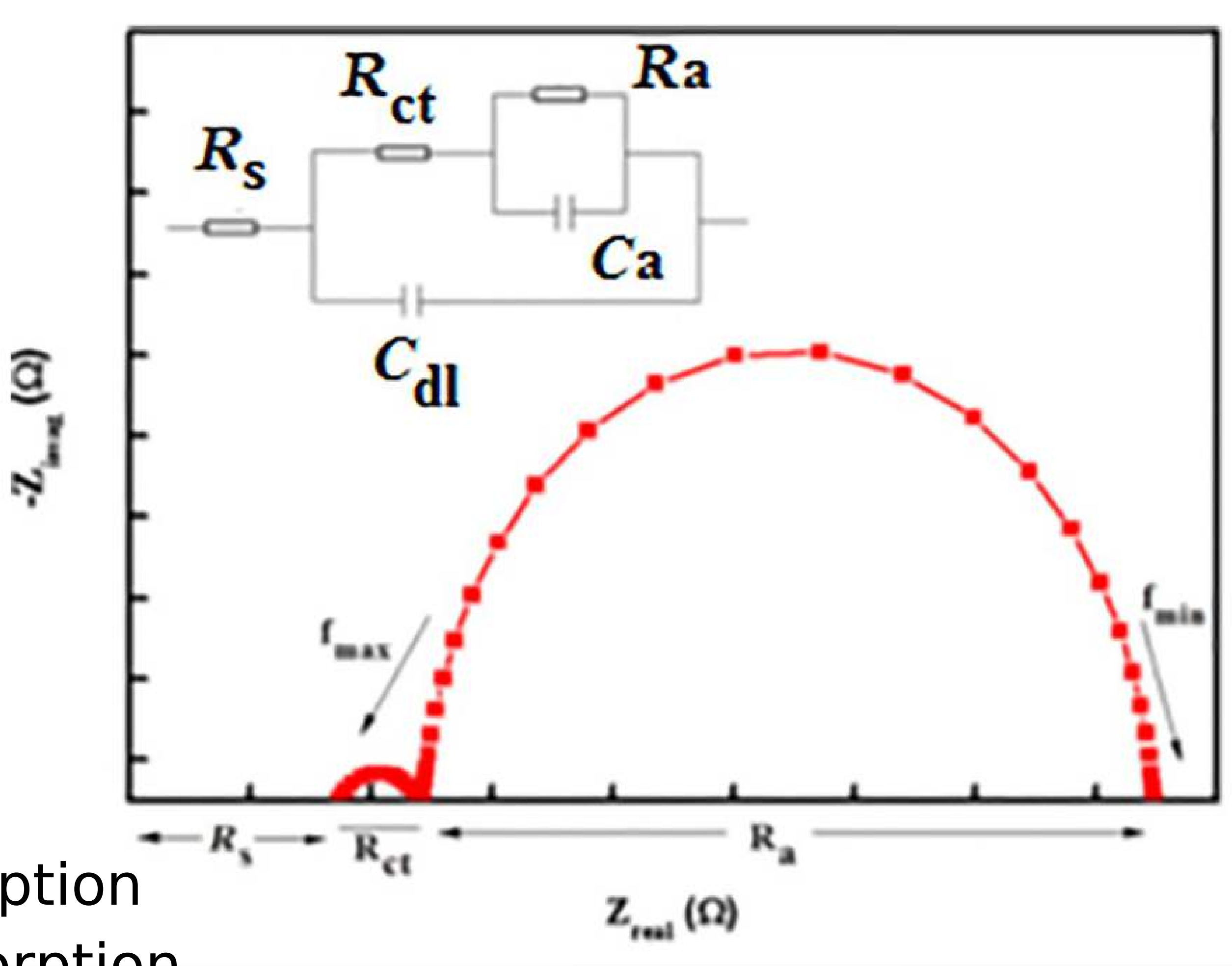
- three–electrode $\Rightarrow$ voltammetry, amperometry
- two–electrode $\Rightarrow$ electrochemical impedance spectroscopy (EIS)



Cyclic voltammogram (A) and amperometry plot (B)

4. Electrochemical Sensors

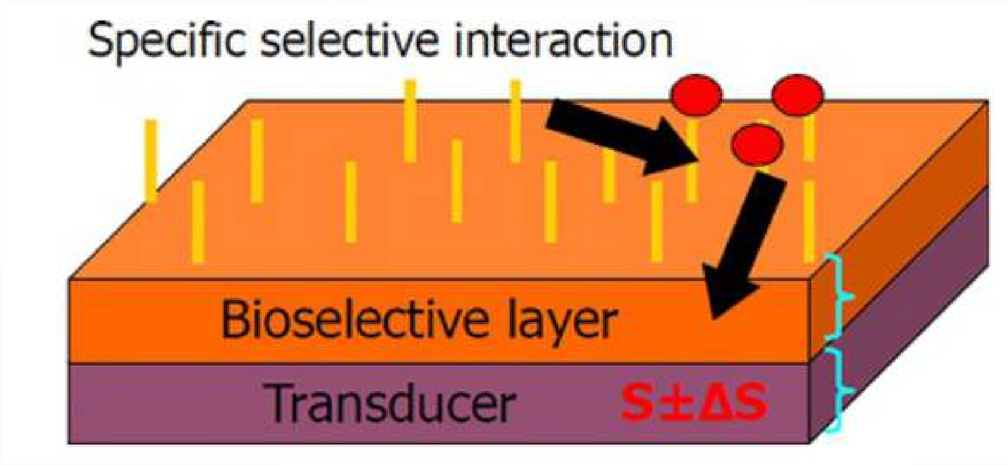
- Schematic representation of Nyquist plot expressing the impedance parameters of the sensor



R_s = solution resistance,
R_{ct} = charge transfer resistance
C_{dl} = double-layer capacitance
R_a = resistance of adsorption/ desorption
C_a = capacitance of adsorption/ desorption

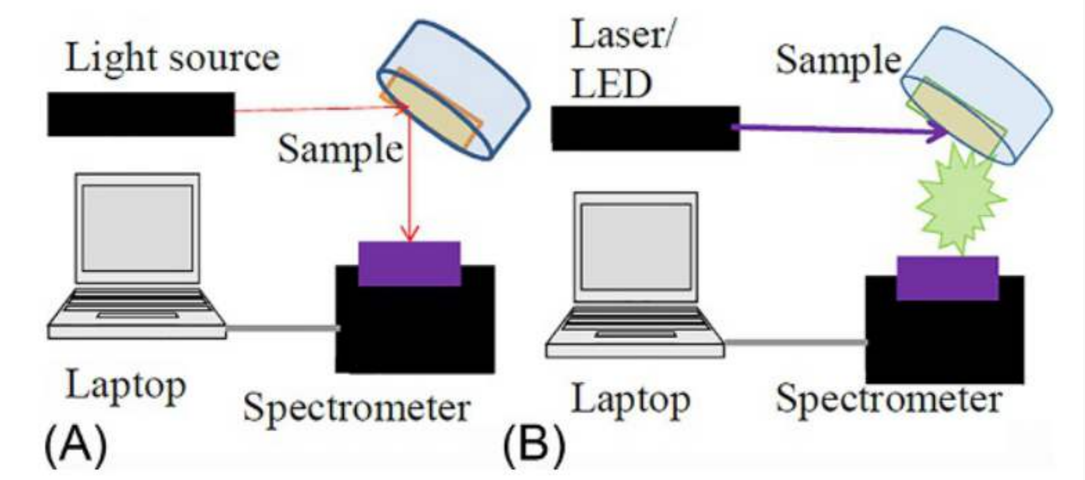
5. Biosensors

- Contains two main parts: (1) bioselective layer and (2) transducer
- Capturing of biomolecules results in the change of parameters of bioselective layer (biological signal) that induces changes of physical properties of the transducer (S+ΔS)
- type of transducers: electrical, electrochemical, optical, magnetic
- types of bioselective layers: enzymatic (interaction enzyme-target molecule), immune (interaction antigens-antibodies), aptamer (DNA), etc.

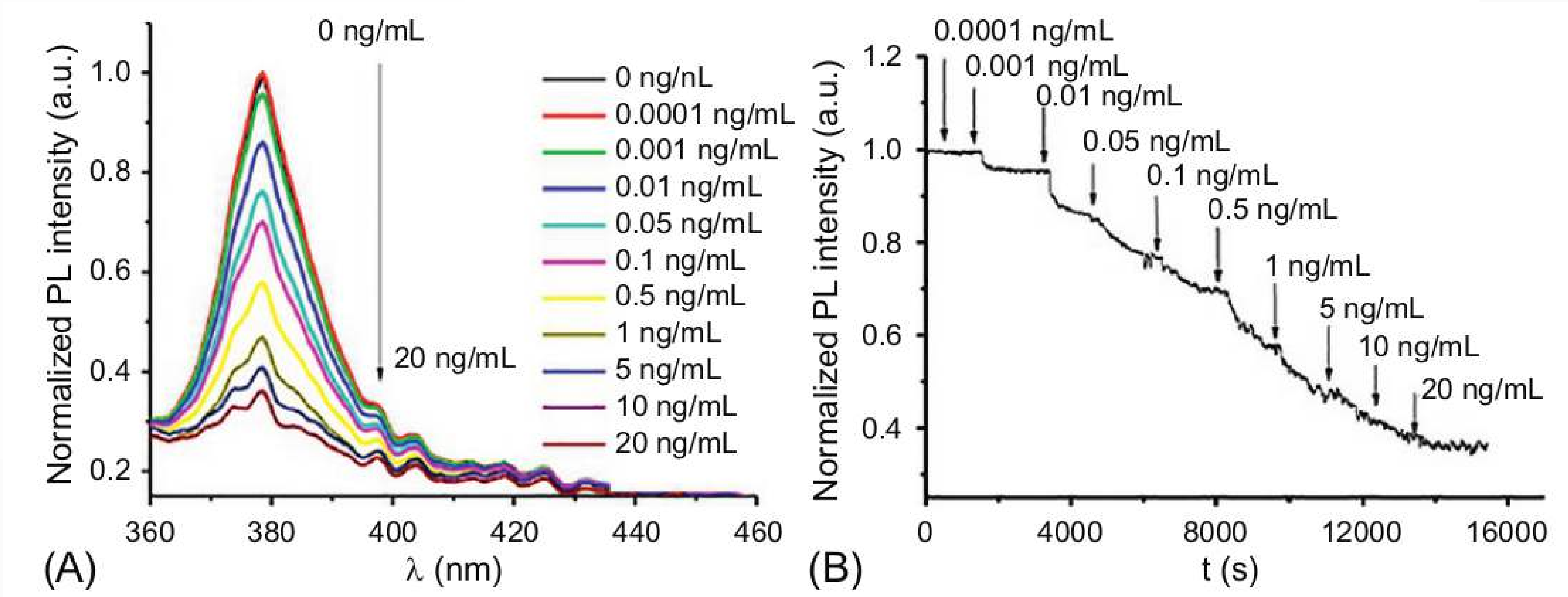


Metal Oxide Optical Biosensors

- The main measurements used in optical sensing are transmittance, absorbance, reflectance, and PL.



Ochratoxin A (OTA) adalah kelas mikotoksin yang diproduksi oleh genus Aspergillus dan Penicillium.



Response of Glass/ZnO-NRs/Protein-A/BSA&Anti-OTA

Metal Oxide Optical Biosensors

Table 2.3 Summary of sensitive properties of optical metal oxide biosensors

Target molecule	Type of biosensor	Transducer	Bioselective layer	Sensitivity range	LOD
Glucose	Reflectance	Au/ZnO	Glucose oxidase	0–300 mg/dL	20 mM
	PL	ZnO		10–130 mM	10 mM
		ZnO:Al		20 µM–10 mM	20 µM
Rabbit immunoglobulin G	Reflectance	TiO_2 nanotubes	Protein A	~0.001 mg/mL	–
Meningitidis DNA	Reflectance	Au/ZnO	ssDNA	10–180 ng/µL	5 ng/µL
Cholesterol	Reflectance	Au/ZnO	Cholesterol oxidase	0.12–10.23 mM	0.12 mM
D-sorbitol	Transmittance	Ag/Ta_2O_5	Sorbitol dehydrogenase	0.1–1.1 µg/mL	3.6 ng/mL
Cortisol	Absorbance	ZnO/ polypirrole	MIP	0–10 to 6 g/mL	26 fg/mL
Ochratoxin A	PL	ZnO	Antibody	0.1–1 ng/mL	0.01 ng/mL
Salmonella	PL	ZnO	Antibody	1000–10^6 cells/mL	100 cells/mL
		TiO_2		10^3–10^5 cells/mL	100 cells/mL
Avidin-HRP	PL	ZnO	Protein A	1–8 µg/mL	–